

An Analysis of Explainable artificial intelligence

Explainable artificial intelligence

Explainable artificial intelligence -- Part 1

Goals Cooperation between agents – in this case, algorithms and humans – depends on trust. If humans are to accept algorithmic prescriptions, they need to trust them. Incompleteness in formal trust criteria is a barrier to optimization. Transparency, interpretability, and explainability are intermediate goals on the road to these more comprehensive trust criteria. This is particularly relevant in medicine, especially with clinical decision support systems (CDSS), in which medical professionals should be able to understand how and why a machine-based decision was made in order to trust the decision and augment their decision-making process. AI systems sometimes learn undesirable tricks that do an optimal job of satisfying explicit pre-programmed goals on the training data but do not reflect the more nuanced implicit desires of the human system designers or the full complexity of the domain data. For example, a 2017 system tasked with image recognition learned to "cheat" by looking for a copyright tag that happened to be associated with horse pictures rather than learning how to tell if a horse was actually pictured. In another 2017 system, a supervised learning AI tasked with grasping items in a virtual world learned to cheat by placing its manipulator between the object and the viewer in a way such that it falsely appeared to be grasping the object. One transparency project, the DARPA XAI program, aims to produce "glass box" models that are explainable to a "human-in-the-loop" without greatly sacrificing AI performance. Human users of such a system can understand the AI's cognition (both in real-time and after the fact) and can determine whether to trust the AI. Other applications of XAI are knowledge extraction from black-box models and model comparisons. In the context of monitoring systems for ethical and socio-legal compliance, the term "glass box" is commonly used to refer to tools that track the inputs and outputs of the system in question, and provide value-based explanations for their behavior. These tools aim to ensure that the system operates in accordance with ethical and legal standards, and that its decision-making processes are transparent and accountable. The term "glass box" is often used in contrast to "black box" systems, which lack transparency and can be more difficult to monitor and regulate. The term is also used to name a voice assistant that produces counterfactual statements as explanations.

Explainable artificial intelligence -- Part 2

Voting Cailloux and Endriss present a method for explaining voting rules using the axioms that characterize them. They exemplify their method on the Borda voting rule. Peters, Procaccia, Psomas and Zhou present an algorithm for explaining the outcomes of the Borda rule using $O(m^2)$ explanations, and

prove that this is tight in the worst case.

Explainable artificial intelligence -- Part 3

Regulation As regulators, official bodies, and general users come to depend on AI-based dynamic systems, clearer accountability will be required for automated decision-making processes to ensure trust and transparency. The first global conference exclusively dedicated to this emerging discipline was the 2017 International Joint Conference on Artificial Intelligence: Workshop on Explainable Artificial Intelligence (XAI). It has evolved over the years, with various workshops organised and co-located to many other international conferences, and it has now a dedicated global event, "The world conference on eXplainable Artificial Intelligence", with its own proceedings. The European Union introduced a right to explanation in the General Data Protection Regulation (GDPR) to address potential problems stemming from the rising importance of algorithms. The implementation of the regulation began in 2018. However, the right to explanation in GDPR covers only the local aspect of interpretability. In the United States, insurance companies are required to be able to explain their rate and coverage decisions. In France the Loi pour une République numérique (Digital Republic Act) grants subjects the right to request and receive information pertaining to the implementation of algorithms that process data about them.

Explainable artificial intelligence -- Part 4

Scholars sometimes use the term "mechanistic interpretability" to refer to the process of reverse-engineering artificial neural networks to understand their internal decision-making mechanisms and components, similar to how one might analyze a complex machine or computer program. Studying the interpretability of the most advanced foundation models often involves searching for an automated way to identify "features" in generative pretrained transformers. In a neural network, a feature is a pattern of neuron activations that corresponds to a concept. A compute-intensive technique called "dictionary learning" makes it possible to identify features to some degree. Enhancing the ability to identify and edit features is expected to significantly improve the safety of frontier AI models. For convolutional neural networks, DeepDream can generate images that strongly activate a particular neuron, providing a visual hint about what the neuron is trained to identify.

Explainable artificial intelligence -- Part 5

Within artificial intelligence (AI), explainable AI (XAI), generally overlapping with interpretable AI or explainable machine learning (XML), is a field of research that explores

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methods that provide humans with the ability of intellectual oversight over AI algorithms. The main focus is on the reasoning behind the decisions or predictions made by the AI algorithms, to make them more understandable and transparent. This addresses users' requirement to assess safety and scrutinize the automated decision making in applications. XAI counters the "black box" tendency of machine learning, where even the AI's designers cannot explain why it arrived at a specific decision. XAI hopes to help users of AI-powered systems perform more effectively by improving their understanding of how those systems reason. XAI may be an implementation of the social right to explanation. Even if there is no such legal right or regulatory requirement, XAI can improve the user experience of a product or service by helping end users trust that the AI is making good decisions. XAI aims to explain what has been done, what is being done, and what will be done next, and to unveil which information these actions are based on. This makes it possible to confirm existing knowledge, challenge existing knowledge, and generate new assumptions.

system and when the users do not have enough trust in the system, they will not exhaust the benefits inherent in it.

Explainable artificial intelligence -- Part 6

Understanding versus trust The goal of explainability to end users of AI systems is to increase trust in the systems, even "address concerns about lack of 'fairness' and discriminatory effects". However, even with a good understanding of an AI system, end users may not necessarily trust the system. In one study, participants were presented with combinations of white-box and black-box explanations, and static and interactive explanations of AI systems. While these explanations served to increase both their self-reported and objective understanding, it had no impact on their level of trust, which remained skeptical. This outcome was especially true for decisions that impacted the end user in a significant way, such as graduate school admissions. Participants judged algorithms to be too inflexible and unforgiving in comparison to human decision-makers; instead of rigidly adhering to a set of rules, humans are able to consider exceptional cases as well as appeals to their initial decision. For such decisions, explainability will not necessarily cause end users to accept the use of decision-making algorithms. We will need to either turn to another method to increase trust and acceptance of decision-making algorithms, or question the need to rely solely on AI for such impactful decisions in the first place. However, some emphasize that the purpose of explainability of artificial intelligence is not to merely increase users' trust in the system's decisions, but to calibrate the users' level of trust to the correct level. According to this principle, too much or too little user trust in the AI system will harm the overall performance of the human-system unit. When the trust is excessive, the users are not critical of possible mistakes of the